

		Reg. No.:	
		Name :	
 			
MID TERM EXAMINATIONS – November 2024			
Programme	: B.Tech.	Semester	: Interim Semester 2024-25
Course Title	: Computer Architecture and Organization /	Course Code	: CSE2003
Date/Session	: 14 Nov 2024 / II	Slot	: C11+C12+C13
Time	: 1 ½ hours	Max. Marks	: 50
Answer all the Questions			
Q.No.	Sub. Sec.	Question Description	Marks
1	a.	Different types of instructions have their own strengths and weaknesses. Explain the trade-offs between using 1-address, 2-address, and 3-address instruction formats in terms of: • Memory usage • Instruction length • Speed of execution • Code complexity	10
2	(a)	Determine the constraints of a 32-bit instruction format used by a processor with 98 instructions and 6 addressing modes. Determine the bit allocation for the opcode and mode fields. Further discuss how the remaining bits could be distributed among the operands, given that instructions can take 0, 1, 2 or 3 operands.	5
	(b)	Analyze and justify why each addressing mode (X: Indirect addressing, Y: Immediate addressing, Z: Auto decrement addressing) is best matched with the given usage categories (1: Loops, 2: Pointers, 3: Constants). Discuss the advantages of using these addressing modes in their specific contexts.	5
3		Add the two floating-point numbers 1.5 and 2.75 using IEEE 754 32-bit floating-point format.	10
4		The following memory units are specified by the number of words times the number of bits per word. How many address lines and input-output data lines are needed in each case? Specify the number of bytes that can be stored in memories. (a) 2K X 16 (b) 64K X 8. (c) 16M X 32 (d) 4G X 64.	10
5		Compare and contrast the power consumption, speed and cost of SRAM and DRAM. Analyze how these factors influence their usage in various types of computing devices.	10
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